40's

WW2 → Alan Turing

Mechanical Computing Machine

John von Neumann

50's

TV, STORAGE

Tranzistor → Logical Gates

(AT&T)

→ Electronical. George Boole

⇒ ENIAC

(XIX, Britain)

f encoding ← MACHINE CODE

f^{-1} decoding ← BYTE CODE

Boolean Algebra

+, -, /, *

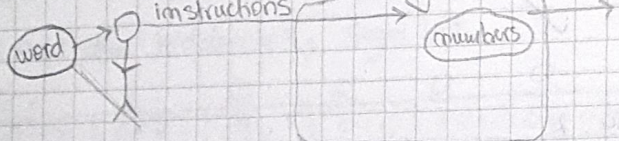
a more effective machine:

Thomas Eddison light bulbs ⇒ tranzistors

Claude Shannon

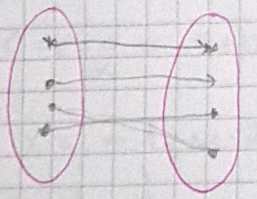
old before Aristotel logic

Programming:



Program: a finite succession of instructions

A × B ↝ cartesian product (Cartesius → René Descartes)



$A \times B \supseteq R \supseteq f \supseteq \nexists bij.$

$\{0,1\} \times \{0,1\} = ?$

$= \{(0,0), (0,1), (1,0), (1,1)\}$

$= \quad < \quad > \quad =$

$\nexists bij. \rightarrow \exists f^{-1}.$

byte: domeniul său algebric: $\underbrace{\{0,1\} \times \{0,1\} \dots \times \{0,1\}}_{8} =$

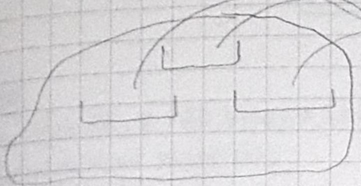
$= 2^8 = 256$

HOV instrucțiune către machine code. (instruction code)

adăugăm fiecărui cod între 0,255 ....

ADD

SUB



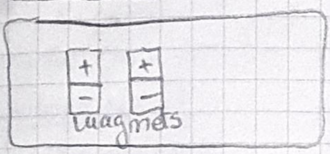
→ formal parameters.

=) CALL PROCEDURE =) return result.

PROCEDURE NAME

storage was the hardware needed to store that procedure.

STORAGE



=) STORED PROCEDURE

→ Procedural Programming paradigm.

↓

Modular programming

(Appollo 11 github) 60' NASA

start:

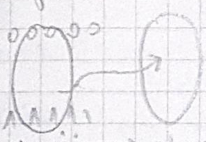
....

call [exit]

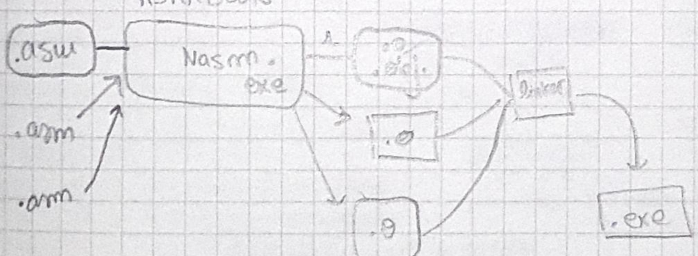
CALL STACK

.asm .agc → text file

FILE = finite succession of bytes.



ASAMBLOR



.h → abstraction

.c → implementation

70's

Grace Hopper → Mark 1 (marine) US army US navy debug

→ Compiler Cobol.

AT&T Bell lab { Dennis Richie, Ken thomson } => C

Data Structures and Algorithms.

UNIX (First OS)

Arpanet

C++

everything is an object.

80's

=) Paradigm Shift.